

Assignment 1: Opinion Network Formation

DPCN Course — 2026

20 September 2026

Team Name

clankers

GitHub Link to Your Code

Repository: <https://github.com/DPCN-M26/Opinion-Network-Formation-A1.git>

Note: The repository contains: `Opinion_Network_Analysis.ipynb` (the full annotated Jupyter Notebook) and `dataset.csv` (the survey data). All plots are generated reproducibly by running the notebook.

1 Dataset Documentation

The dataset contains survey responses from **96 students**, each answering **60 questions** (15 per topic) on the following four categories:

Code	Topic	No. of Questions
T01–T15	Technology & Artificial Intelligence	15
E01–E15	Education & Academia	15
S01–S15	Society & Ethics	15
V01–V15	Environment & Sustainability	15

Each question was answered on a **5-point Likert scale**: Strongly Disagree, Disagree, Neutral, Agree, Strongly Agree. A small number of responses were marked "No Comments."

Preparation & Encoding

1. **Numerical mapping:** We mapped responses to signed integers to capture opinion direction: Strongly Disagree $\rightarrow -2$, Disagree $\rightarrow -1$, Neutral $\rightarrow 0$, Agree $\rightarrow +1$, Strongly Agree $\rightarrow +2$. "No Comments" was mapped to 0 (treated as Neutral/abstention).

2. **Baseline sentiment:** The average scores show that the class held *consistently positive* views across all topics:

- Technology: **+0.928** — Education: **+0.819**
- Society: **+1.151** — Environment: **+1.226**

Environment and Society questions received the strongest agreement overall, while Education responses showed the most nuanced spread.

3. **Positive-shifting for TF-IDF:** To apply TF-IDF (which requires non-negative counts), the values were shifted from $[-2, +2]$ to $[1, 5]$, preserving relative magnitudes.

2 Pipeline Followed

Our pipeline is innovative because it goes **beyond standard cosine similarity**. We use three complementary approaches that are central to the theme of this course: *network construction*, *static network analysis*, and *dynamical processes on networks*.

2.1 Step 1: TF-IDF Weighted Similarity Network

Instead of treating all shared opinions equally, we applied **Term Frequency-Inverse Document Frequency (TF-IDF)** — borrowing from Natural Language Processing — where each student is a "document" and each survey response is a "term." TF-IDF assigns a lower weight to responses that the *majority* of the class agreed on, and a *higher weight* to rare, distinguishing opinions. This ensures that the resulting edge weights reflect **genuine ideological proximity**. After computing pairwise cosine similarity, we applied a **70th percentile threshold** to retain only the strongest 30% of potential connections.

2.2 Step 2: Community Detection (Louvain Algorithm)

We applied the **Louvain algorithm** (modularity-maximization) to detect communities within the TF-IDF network. This algorithm partitions nodes into communities that maximize intra-community edge density relative to chance (measured by the **modularity score Q**).

2.3 Step 3: Multiplex Network Analysis (4-Layer)

We constructed a **4-layer multiplex network** — one layer each for Technology, Education, Society, and Environment — to study whether opinions are correlated *across topics*. A high Jaccard edge-overlap between two layers means students who agree ideologically on one topic also tend to agree on another.

2.4 Step 4: Opinion Dynamics — Deffuant Bounded Confidence Model

We simulated how classroom consensus (or fragmentation) would emerge if students interacted and influenced each other using the **Deffuant model**:

- Initial opinion state = average score across all 60 questions.
- At each time step, a random connected pair (u, v) is selected. If their opinions differ by less than the confidence threshold $\varepsilon = 0.5$, they update toward each other: $x_u \leftarrow x_u + \mu(x_v - x_u)$, with convergence rate $\mu = 0.3$.

- The simulation runs for **8,000 steps** on the TF-IDF network.
- A GIF animation (`opinion_dynamics.gif`) visually shows how node colours (red=disagree, green=agree) shift across the network over time.

3 Analysis and Visualizations

3.1 Network Metrics Summary

Metric	Value
Nodes (students)	96
Edges (strong ideological links)	1,368
Network Density	0.30
Average Degree	28.50
Average Clustering Coefficient	0.678
Connected Components	11
Communities (Louvain)	13
Modularity (Q)	0.106

The network has a **density of 0.30** (moderately dense) and a high **average clustering coefficient of 0.678** (strong clique structure). The **11 connected components** reveal that a small number of students are ideological outliers who share strong connections with no one in the main cluster.

3.2 Centrality: Opinion Influencers

We compute two complementary centrality measures:

Degree Centrality (broadest connectors):

Rank	Student	Degree Centrality
1	Student 44	0.7368
2	Student 60	0.7368
3	Student 68	0.7368
4	Student 73	0.7368
5	Student 78	0.7368

Betweenness Centrality (opinion brokers bridging communities):

Rank	Student	Betweenness Centrality
1	Student 67	0.0533
2	Student 14	0.0249
3	Student 77	0.0242
4	Student 97	0.0237
5	Student 41	0.0235

Student 67 is the most critical structural actor: the key ideological **broker** between communities.

3.3 Figure 1: TF-IDF Opinion Network

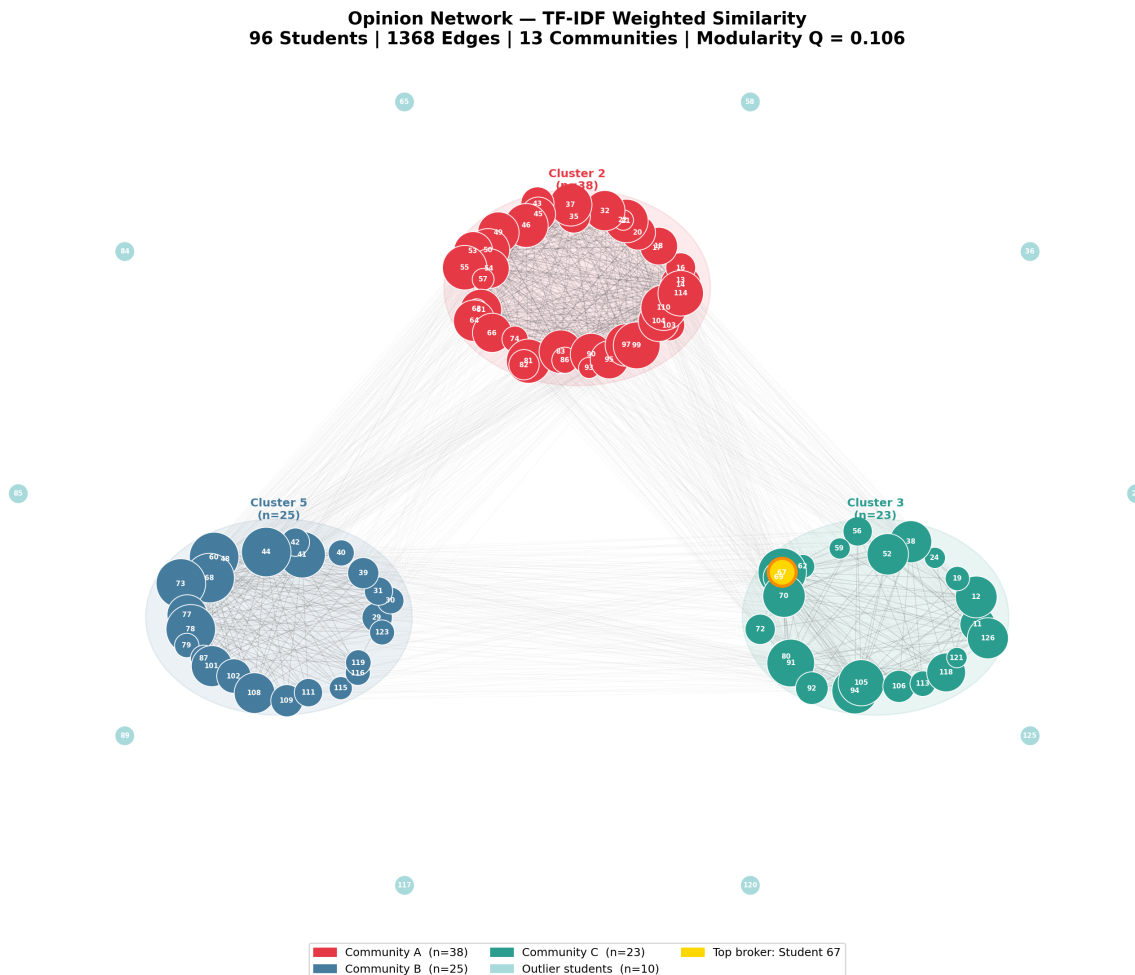


Figure 1: The Opinion Network arranged using a community-aware layout. The three main clusters are placed at the vertices of a triangle. Node size scales with degree centrality. The gold-ringed node is **Student 67** (highest betweenness centrality). The 10 outlier students (cyan) float around the periphery.

The network reveals three dominant ideological clusters (n=38 red, n=25 blue, n=23 teal) accounting for **86% of the class**. This confirms the class is **not polarized** (no two opposing camps), but differentiates into three ideologically nuanced groups with significant common ground.

3.4 Figure 2: Topic-Level Average Scores

Almost all bars lean positive. Within Technology, questions about AI replacing jobs (T06) and government regulation (T12) show the lowest scores. Within Education, opposition to mandatory attendance (E03) and skepticism about traditional exams (E02) stand out.

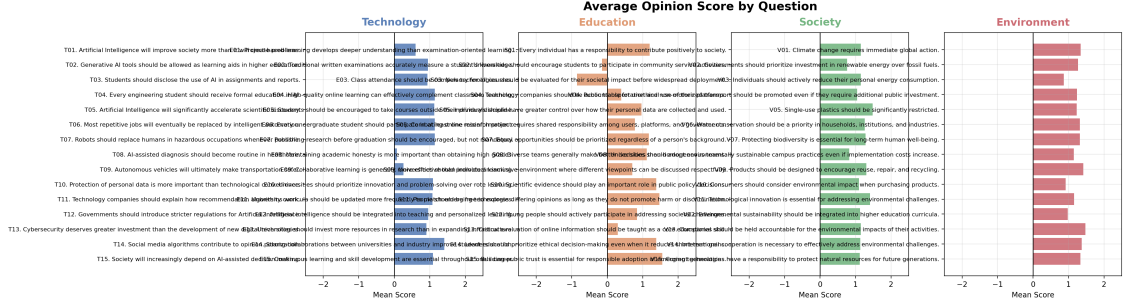


Figure 2: Mean response score (-2 to $+2$) per question, broken down by the four survey topics.

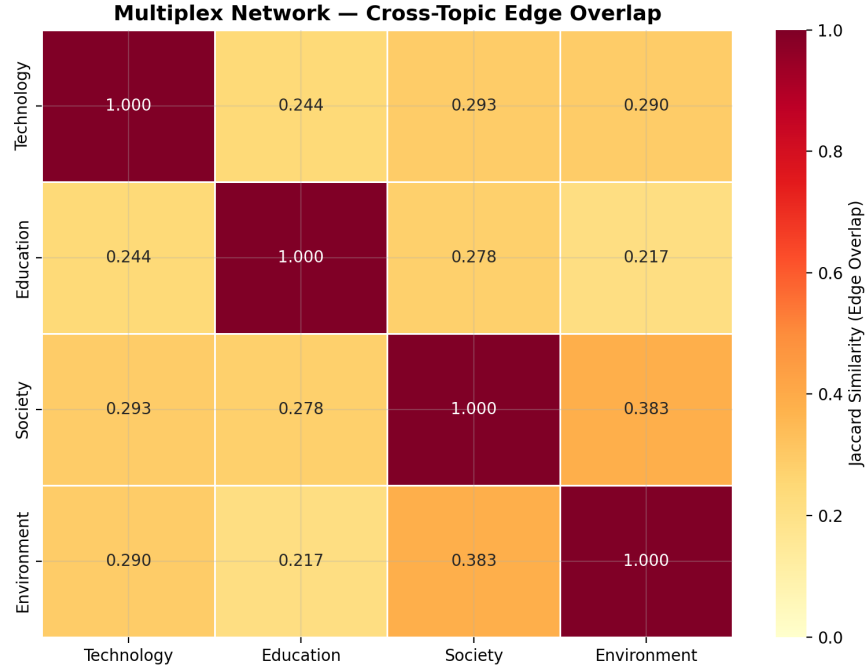


Figure 3: Jaccard similarity of edge sets between the four topic-specific network layers.

3.5 Figure 3: Multiplex Edge Overlap

The highest overlap is between **Society** and **Environment** (**0.383**). The weakest connection is between **Education** and **Environment** (**0.217**), suggesting that educational philosophy is the most independent of all four topic domains.

3.6 Figure 4: Opinion Dynamics (Deffuant Model)

- **Initial distribution:** Mean = $+1.03$, spread widely across $[-0.2, +1.8]$.
- **Final distribution:** Mean = $+1.03$, Std = 0.376 , fragmented into **8 distinct consensus bands**.
- The system does **not** converge to a single consensus. Instead, it freezes into **8 attractor states** (echo chambers), the largest containing 72 students. The 10 outlier students retain

Deffuant Bounded Confidence Dynamics on the TF-IDF Opinion Network

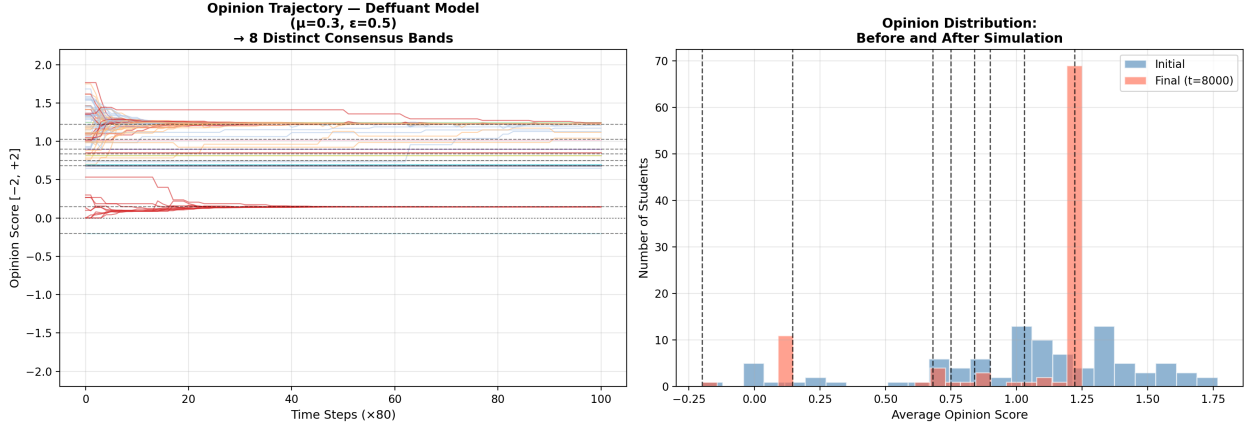


Figure 4: (Left): Opinion trajectory for each student over 8,000 simulation steps. (Right): Histogram showing the dramatic narrowing of the initial spread into 8 distinct attractor bands.

their initial opinions, as they have no network connections — a direct demonstration that **topology constrains dynamics**.

4 Results and Discussion

The analysis reveals several key insights:

1. **The class is not polarized; it is differentiated into a nuanced consensus.** Three large clusters account for 86% of students.
2. **Opinion alignment is topic-specific, not global.** The multiplex analysis proves that someone who shares your views on Technology is not necessarily aligned with you on Education.
3. **The most structurally important student is not the most popular one.** Student 67 has the highest *betweenness* centrality, while others have the highest *degree* centrality.
4. **Bounded confidence creates echo chambers, not consensus.** The Deffuant simulation shows the system fragmenting into 8 final attractor states.
5. **Network topology directly controls which dynamics are possible.** The isolated outlier students never change their opinions because they have no edges in the network.

5 Individual Contribution

Team Member	Tasks
Ch shreyash	End-to-end pipeline design, data preprocessing, network construction (TF-IDF), community detection, multiplex analysis, Deffuant model simulation, visualization programming, and report drafting.
Vansh Motwani	Codebase review, verification of analysis methodology, parameter tuning validation, and peer-review of the final report.